

The Nautilus Bundle Manual

V2.1.0

© Audio Ease 2004

System requirements:

Macintosh G4/400 MHz or better
MACOS 10.2.3 or higher

Available in the following formats:

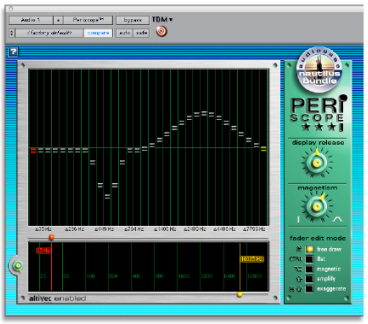
HTDM/RTAS/AudioSuite (for Pro Tools TDM/LE systems)
MAS (for Motu Digital Performer and Audio Desk)
VST (for Cubase and Nuendo & others)
Audio Unit (for Logic & others)

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Real time Granular Synthesizer

★★★ RIVERRUN

Page 8 — 

PERI SCOPE ★★★

32 band zoomable phase correct equalizer and spectroscope

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DEEP PHASE NINE

24 notch syncable true deep phaser

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Purchase & Authorization, getting rid of the demo beeps. Page 3



An unauthorized version of the Nautilus Bundle beeps every few seconds. To authorize your copy:

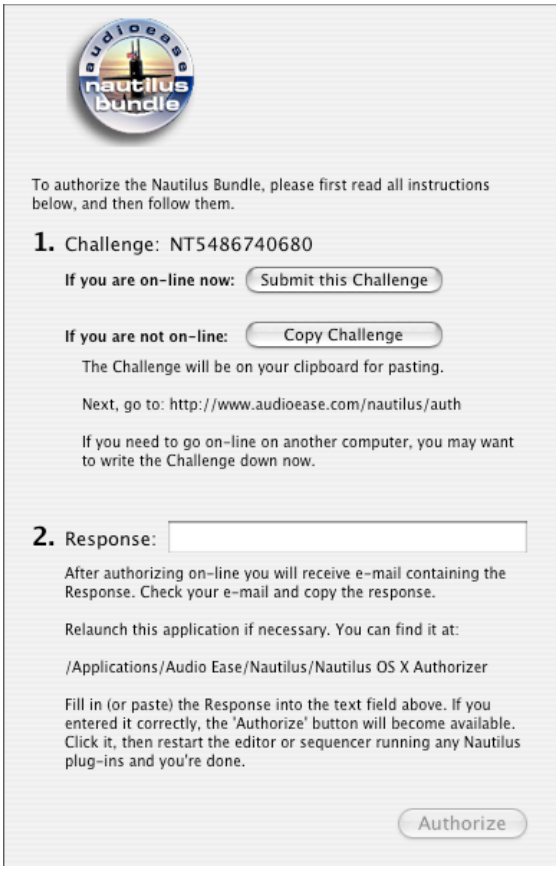
1. Install the Nautilus Bundle release version from CD or a download.

2. The installer automatically launches the Nautilus OS X authorizer.

3. In the authorizer, you see the challenge code generated for your system.

4a. If the computer you're installing Nautilus on is not connected to the internet, write down the challenge code and web address quoted and go to a computer that is connected to the internet.

4b. If the computer you're installing Nautilus on is connected to the internet, press the button labeled 'Submit this challenge'



5. On the web authorization page, follow the instructions given there.

6. If your authorization was approved, you'll receive an e-mail with the response code.

7. If the Nautilus OS X Authorizer is not running, launch it from /Applications/Audio Ease/Nautilus/Nautilus OS X Authorizer

8. If you have a sequencer/editor running with the Nautilus Bundle active, you'll have to quit and relaunch that to make the demo beeps go away.



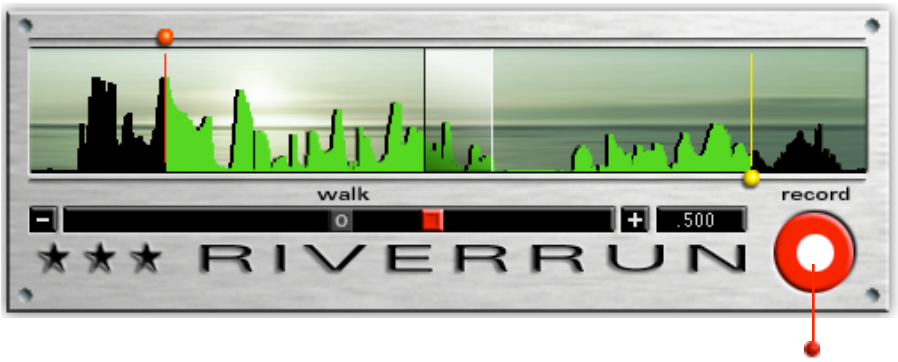
Riverrun, by the maker of Thonk, is the most versatile real time granular synthesizer available today. Audio from your host's tracks or directly from a disk file can be streamed and frozen into Riverrun. Tiny particles of the audio signal called grains are first pushed through an envelope generator, pitched and scattered across the output channels according to the surface controls. There is no synthesis technique that sounds quite like Riverrun's granular synthesis, so do check out the immense Gregorian monk choirs that can be created out of a single utterance. Or use Riverrun as a microscope as the grain cursor progresses through the snapshot at a hundredth of normal speed, dissecting a single guitar strum into a wealth of distinct stages. With the rhythmic mode it is possible to groove random samples locked to your editor's tempo, while pitches are shifted to pre- selected chords.
Most of Riverrun's parameters can also run under automatic control.

The Nautilus Bundle has extensive balloon help.
Turn it on and off with this toggle.

The input panel indicates where the grains come from at any given point in time.

The Grain Glass in the input screen indicates the portion of audio that the currently audible grains originate from. The position of the Grain Glass is controlled by the 'Walk' slider. When the Walk slider is set to a positive value, the Grain Glass moves through the input screen from the left (red) needle to the right (yellow) needle. The higher the Walk value, the faster the Grain Glass moves. It can also move backward, and you can move the Grain Glass by simply clicking and dragging anywhere between the red and yellow needles.

The size of the Grain Glass, and thus of the grains that you hear, is controlled by the grain length slider that is discussed below.



When the host is playing and the record button is depressed, audio from the channel that Riverrun is inserted in streams into the input screen, from right to left. Press record again and the audio in the input screen freezes.



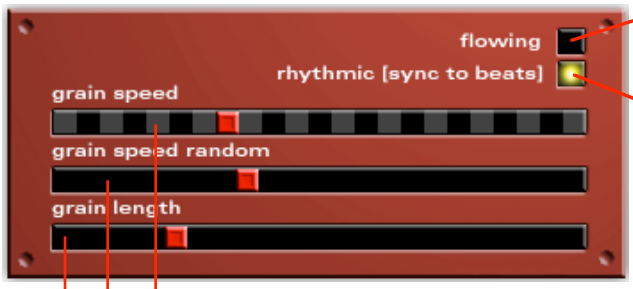
The load button allows you to load Sounddesigner II(tm) audio files directly from disk into the input screen. The save button allows you to store the current contents of the input screen to disk as a Sounddesigner™ II file.

When you close your song or session, Riverrun automatically saves the contents of its input screen as a Sounddesigner™ II file in a folder called "Riverrun Snippets". This folder sits in the folder '/Library/Application Support/Riverrun Scratch *f*'. This procedure allows Riverrun to show up exactly the way you had left it when you reopen a song. This automatic saving only happens when you have touched the Record button. If you have fed the input screen via the load button, and have not touched record, no file is saved to disk on close. When the song is reopened your previously selected file is reloaded.

Another exception is when you have just saved the contents of the input screen via the save button. If you have not touched Record after you have saved, no file will be saved to disk on close. When the song is reopened your saved file is reloaded.

The location of a snippet is stored into your host's presets. When you select a preset, the accompanying file will be loaded into the input screen.

The Texture panel offers control over the fluidness and density of the output grain stream



The flowing mode is designed to create dense textures, while the rhythmic mode allows you to create beat patterns locked to the host's tempo.

When Rhythmic is selected, gray squares appear in the Grain Speed Slider, and the thumb sticks to these gray and black fields. The black fields represent even divisions of the host's tempo such as whole , half, 1/4 notes down to 1/128 notes. The gray fields represent the triplets in between.

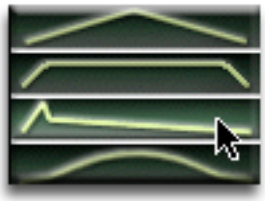
The Grain Speed slider controls the amount of grains that are taken from the input screen. Together with the Grain Length slider it determines the density of your output grain stream. When you allow Riverrun to pretty much take over your entire processor by setting both grain speed and grain length to their upper limits, the output grain stream can be up to 1000 layers thick.

If Grain Speed Random is set low, the grains appear at regular time-intervals in the output grain stream and you can often hear the repetition as a steady tone, the pitch of which is set by Grain Speed. However, if you set Grain Speed Random to a higher value, the grains break loose from their calculated output positions and start to wander around. The output texture loses the grain pitch and becomes much more chaotic and organic. In Rhythmic mode Grain Speed Random tends to humanize the beats up to the point of total anarchy.

With the Grain length slider you determine the size of the grains that are taken from the input screen. When you drag the slider, you'll see the Grain Glass' size change accordingly. When the Grain length slider is set all the way to the left grain lengths are set to mere milliseconds, while all the way to the right grains are half a second long.

"shape" is an envelope that is superimposed on every grain taken from the input screen. There are flowing and punchy ones for various effects.





When the mono/stereo slider is set to mono, all output grains are sent to dead center of the output grain stream. In this mode the pitch that results from the grain repetition becomes most apparent.

When set to stereo, every grain ends up in a randomly determined place between the far left and right of the stereo panorama. Slowly moving the fader from mono to stereo will give the effect of 'opening up' the grain stream.

When using Riverrun with Quad or 5.0 outputs, the mono/stereo slider has the same effect over all outputs: so setting it to mono would place the position in the middle, roughly your listening position, set to stereo it the grains are randomized over all speakers.

All controls in the Transposition Panel together determine the transposition of each grain.

To get a feel for these settings use the Grain Glass to select a nice piece of a 'pure note' such as part of a sung vowel or a single piano note. You may want to halt the Grain Glass by clicking in the little zero in the 'Walk' Slider.

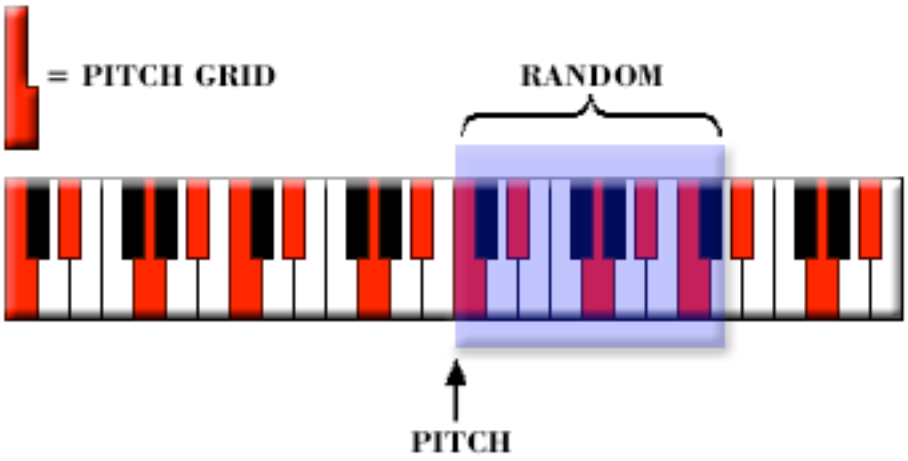


When a grain is taken from the input screen, and provided with an envelope from the shape popup it is time for Riverrun to figure out what its transposition should be.

Riverrun takes the Pitch Grid, Pitch, and parameters, throws the dice, and picks a single pitch for each output grain. At the end the pitch is passed through Fine Tune.

When Pitch Grid is set to Seamless (no grid), the Pitch parameter tunes smoothly.

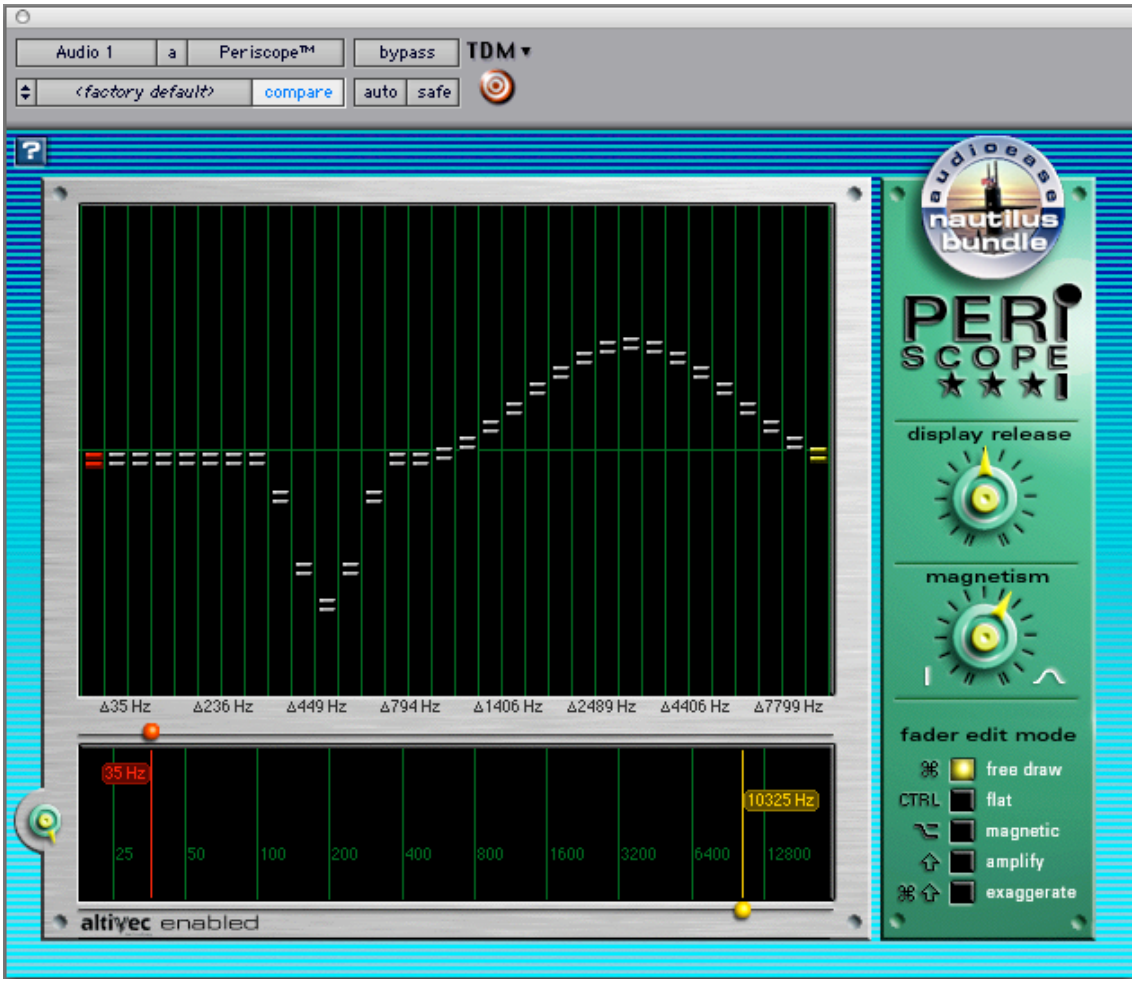
The figure below shows how the parameters affect Riverrun's pitch choice.



Example: 'c minor 7th' is chosen as a Pitch Grid. This Pitch Grid contains all the red keys. The *size* of the blue area indicates the Random setting, while the *start* of the blue area indicates the value of the Pitch knob. As a result the pitches for the output grains are randomly picked from the red keys *within* the blue area. Finally Riverrun alters the lucky pitch according to the Fine Tune Setting.

Multichannel usage:

Riverrun has, in the appropriate hosts, multichannel output capabilities. It comes in Stereo, Quad and 5.0. Note that all of these have only mono or stereo input, so Riverrun cannot be inserted in a channel that has Quad or 5.0 inputs. As an example, to use Riverrun stereo-to=quad, create a stereo track (note that, when creating a 'stereo track' in either Pro Tools or Digital Performer, the stereo part pertains to its inputs). Assign the output of this track to a quad bundle. Insert Riverrun in this track. If the quad bundle is a subset of a larger, 5.1 bundle, the outputs will be mixed to the appropriate outputs by the host.



Periscope is a flexible and phase coherent 32 band equalizer that offers incredibly sharp +36 to -144 dB bands. Periscope provides enough resolution for 24 bits audio.

The really amazing thing is that behind your faders a real-time frequency analysis is displayed so you can grab your audio at the exact point where you see it. Push up a fader and you'll not only hear but also see that band come up accordingly. With the needles in the overview monitor you can set the lowest and highest band to any desired frequency, so you can easily zoom in on a frequency area and obtain exceptional band resolution and separation.

Distribute 32 bands from 0 Hz to 400 Hz for powerful control over the low-end of your mix while everything remains completely phase correct. Or surgically fully eliminate just a 10 Hz band at 3 kHz from full bandwidth material. Until you hear it for yourself you won't believe the effects you can create with an equalizer that attenuates and amplifies as drastic as Periscope.

As all faders and outside bands are automatable in Cubase, Nuendo, Digital Performer and Pro Tools, all your movements within the fader view can be recorded and replayed. Also, all hosts provide a way to minutely edit the automation data in ramp overviews of an audio track.

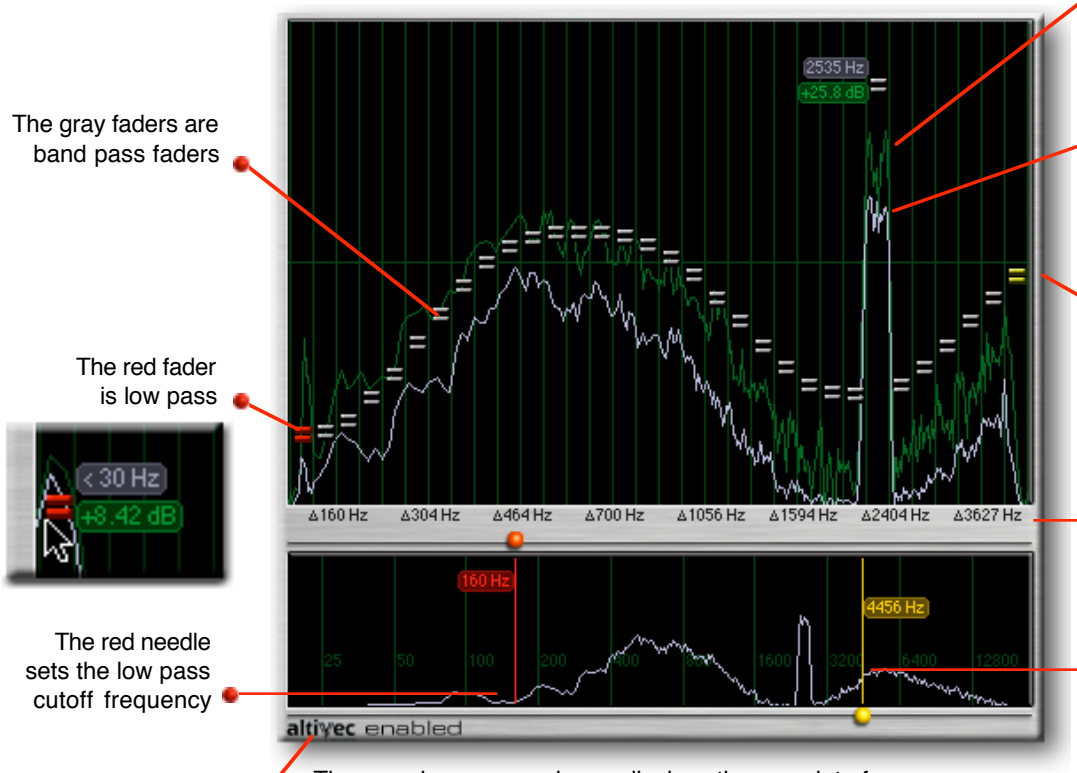
Note that a major difference between recording automation in a regular plug-in and in Periscope, is that the fader view in Periscope always overrides all fader automation, so overdubs are not possible.

Periscope is capable of multichannel work: you can insert it on a mono, stereo, quad, 5.0 or 5.1 track.

The Nautilus Bundle has extensive balloon help. Turn it on and off with this toggle.

The upper screen contains 32 faders. The leftmost red one is a low pass fader, the yellow rightmost fader is a high pass fader. All faders in between are band pass faders. The low and high pass frequencies are determined by the red and yellow needles in the overview window at the bottom of the display. When you drag the needles, the boundaries of the fader screen change and a different piece of the frequency range is displayed and processed. This enables you to create a smooth filter, the cutoffs of which can be manipulated by the needles.

The fader area displays and processes the frequency range between the low and high pass cutoffs set by the red and yellow needles.



The green graph is a peak-type graph

The gray faders are band pass faders

The blue graph is an average graph, the speed of which can be set with the 'display release' knob

The yellow fader is high pass

The red fader is low pass

These are frequency indicators that change while you touch the red and yellow needles

The red needle sets the low pass cutoff frequency

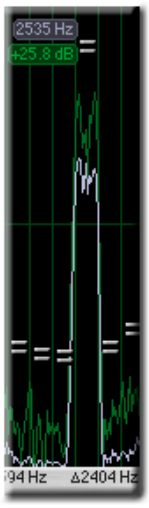
The Yellow needle sets the high pass cutoff frequency

The overview screen always displays the complete frequency range.

This indicates if the Altivec Processor in your G4 is used for Periscope. In fact this indicates if you have a G4 (duh). As a G4 is the minimum requirement for the Nautilus Bundle, this logo will always show.



When a fader is moved, labels popup that display its center frequency and amplification. The labels fade away a few seconds after you release the mouse button





The little knob at the left side of Periscope sets the brightness of both the green peak graph and the green straight lines in the displays.

Periscope is both phase correct and very extreme. Drawing a fader to the bottom of the screen causes the band to completely vanish from the audio, while the adjacent bands stay totally intact, and everything stays phase correct. You can build 'brick wall' type filters this way, or you can surgically remove only that TV-monitor beep from a recording. Alternatively very sharp peaks are possible as well.



In Free Draw mode you can drag the mouse across the fader screen and all faders will snap to the position of the arrow cursor



In Flat mode it does not matter what fader you are operating, you will be dragging all faders up and down simultaneously, in a horizontal line. When you set the faders to the thin green line in the fader screen, the zero dB indicator, the output is the same as the input of Periscope.



In Exaggerate mode, pushing a fader up will cause all faders to move up by the same factor. If you pull down a fader to half its previous height, all faders will be drawn back to half their previous heights. Pushing a fader up will exaggerate the curve, while pulling a fader down will tend to flatten it.



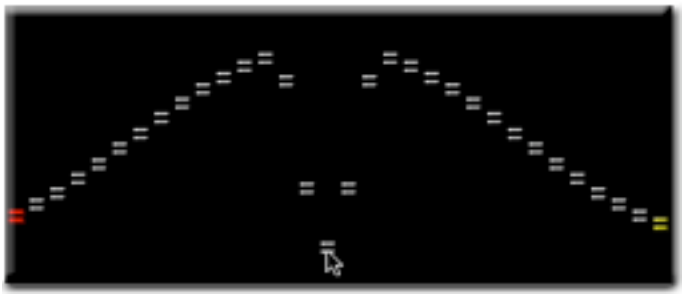
In Amplify Mode, the curves shape does not change, but will just move up and down according to the fader you are operating. The audible effect is that of an overall gain change.

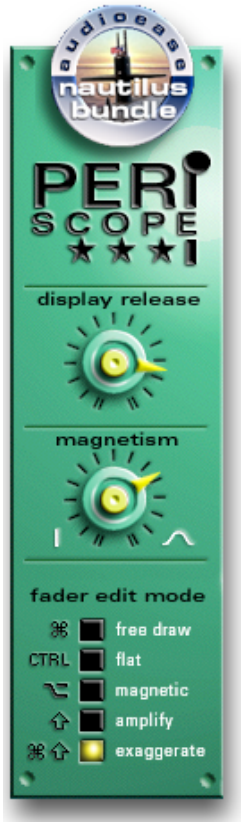


In Magnetic mode, you can smoothly transform the current fader positions by dragging a single fader. The 'Magnetism' knob determines exactly how smoothly. In the example below the center fader is pushed up with a high magnetism setting:



Then the magnetism knob is set quite a bit lower, and the center fader is pulled down again :

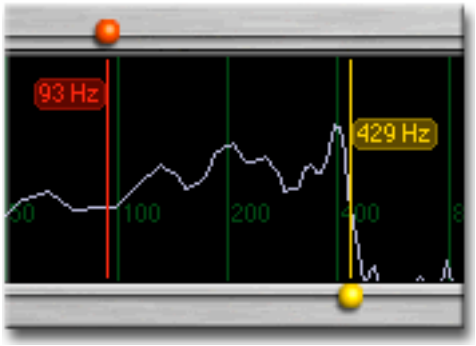




The Display Release sets the amount of time the blue graph uses for its averaging. When set low, the blue (average) graph is as quick as the green (peak) graph. When set high the blue graph reacts very slowly and smoothly to change

The Magnetism knob only has an effect in the magnetic fader edit mode (see left)

The fader edit modes can be selected by clicking in a box, or by key modifiers on your keyboard. When a key modifier (i.e. shift) is released, the fader edit mode snaps back to the setting that was last clicked in.

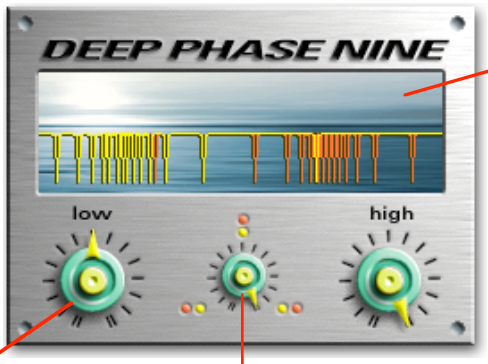


Periscope tries to distribute the frequencies of the faders with a constant relative bandwidth (a constant musical interval). Because of technical reasons the bandwidth cannot be less than approximately 11 Hz. This means that with some settings, the low bands are equally spaced, and the higher have a constant relative bandwidth. When moved, the needles push each other away to maintain space for 30 different bands, automatically deciding where to switch from constant bandwidth to constant relative bandwidth.



Deep Phase Nine is a true phaser (not a flanger) that offers unprecedented control and visual feedback. Up to 24 notches per channel can be swept by a versatile LFO while they are plotted in real time. Peaks and notches can be boosted to the point of self oscillation because a fitted look-ahead-peak-limiter constantly guards the output levels. Automatable one-shots, glassy sample-and-hold phasers that are locked to performer's tempo, deep classic 6 stagers, notch spread control, this is truly a phaser done the Audio Ease way.

Deep Phase Nine v2 has multichannel capabilities. You can use it to make your mono source quadraphonic, insert it on a 5.1 track or master. In that case the small stereo widening knob will diverge the phaser over all output channels.



The notches display panel does its best to reflect all parameters in one single animated graphic. Here we are looking at a stereo phaser. Both channels have sixteen sharp notches, or dips. The yellow notches are in the left channel, most of them in the lower half of the frequency spectrum. The red notches are in the right channel and are offset some distance from the left channel notches by the stereo phase difference knob in the middle of the panel

Normally the notches will be swept according to the LFO settings (explained below). The upper and lower frequency limits of the sweeps are set by the 'low' and the 'high' knob.

If a stereo-to-stereo, or mono-to-stereo Deep Phase Nine is chosen, there will be a small knob in the middle of the notches display panel. This knob determines the offset the right notches have from the left notches. When the knob points upwards, the right and left notches overlap. In Quad, 5.0 and 5.1 modes, this sets the offset between the notch displacement of all channels.

The Nautilus Bundle has extensive balloon help. Turn it on and off with this toggle.

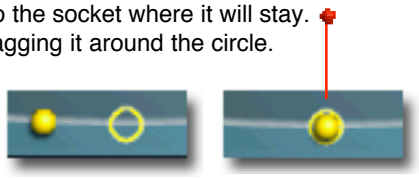
The LFO Panel controls and displays the way the phasers notches sweep through the frequency spectrum. Here you can select an LFO shape, manually control the phase by dragging the yellow ball in the LFO Oval and sync to the host's meter.

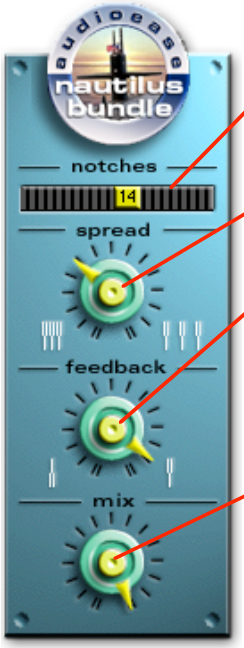


- The LFO Speed Slider is divided into beat multipliers when 'sync to beats' is on. When you select 1/4, it takes the yellow ball four of Performer's beats to complete a full cycle. The sliders thumb also snaps exactly in between the divisions, creating triplets. When the option key is held down you can slide more smoothly, selecting more exotic divisions.
- When sync to beats is switched off, the LFO Speed Slider becomes black and you can smoothly select LFO speeds between 1/256 and 16 Hz.

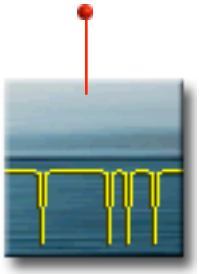


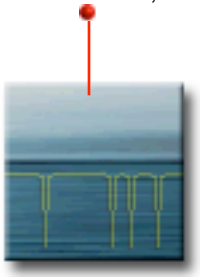
- The sweep of the notches is smoothest with the Sine LFO shape.
- The Triangle shape takes much sharper turns.
- In sample and hold mode random positions are chosen. This effect is particularly interesting when sync to beats is enabled.
- When either of the one shot LFO shapes is selected, you get a ring called the socket. You can now click anywhere on the oval and the ball will travel from the cursor right down to the socket where it will stay. You can position the socket by dragging it around the circle.





- The amount of notches per channel is selected with the notches slider. A classic analog 6 stage phaser gives 3 notches. Deep Phase Nine can go as high as 24 notches per channel.
- You can control the distance between the individual notches with the spread parameter.
- Within Deep Phase Nine, up to 48 so called second order all-pass filters create the notches or peaks. The amount of filter output that is fed back to the inputs determines the sharpness of the peaks and notches. When feedback is negative, you will hear (and see) peaks, when feedback is positive, there will be notches. Very high and very low settings will create whistling (self-oscillation) effects that are kept from getting out of hand by Deep Phase Nine's built in look ahead peak limiter.
- The mix parameter controls the balance between dry (input) and wet (processed) sound. When the knob is set high, the notches in the display are opaque. At a low Mix value, the notches are transparent





Pro Tools notes.

You can use the Nautilus Bundle in Pro Tools TDM system as an HTDM plug-in. This means the plug-in appears to you as regular TDM plug-in, but in the background does its DSP work on the host computer. An HTDM plug-in has some system dependent latency. With a standard Pro Tools installation, this latency is 1028 samples. You can get this latency smaller by installing the '_128' or low latency version of the Digidesign Stream Manager. Your latency will then be 260 samples. Note that the lower latency version of the Stream Manager affects all other HTDM and DirectConnect plug-ins. As always, the lower latencies you use, the higher the processor load. You have to make sure that your system is up to the job before installing the low latency version of the Stream Manager. We do not recommend the lower latency version on systems slower than 1 GHz.

The RTAS version is meant for users of Pro Tools LE systems.

As of version 3.1.0, the Nautilus Bundle can also be used as AudioSuite plug-ins. Note that the AudioSuite plugs appear as distinct mono-mono and stereo-stereo plug-ins: you must choose the proper one for the selection you wish to process in your session or region bin.

Digital Performer notes.

There is only one specific issue with the Nautilus Bundle and Digital Performer. As Digital Performer loads both its own MAS format plug-ins as well as Audio Units, when both versions of the Nautilus Bundle are installed, both will show up. The Audio Unit set will be labeled Periscope AU, Deep Phase Nine AU and Riverrun AU, where the MAS versions will carry their straight names. In this way you can always make a distinction between one and the other.

In general, we recommend using the MAS version only in DP, and not have the Audio Unit installed. The Audio Unit version does not support automation in DP, where the MAS version does.

Cubase & Nuendo notes.

When using any of the Nautilus Bundle plug-ins in Cubase or Nuendo's master channel, you have to understand the routing window that the host provides to indicate the actual channel configuration how the audio is processed. For instance, Riverrun comes in Quad and 5.0 output versions, but does not have that many inputs. In the routing window of the master channel you can see what inputs of Riverrun are connected to what outputs. See the documentation of Cubase or Nuendo for more information on this window.

Logic notes.

To use the Nautilus Bundle in Logic, you need to install the Audio Unit version.

New in versino 2.1.0 is the ability to automate the plug-ins' parameters. Follow the instructions in the Logic documentation on the use of automation.

Also, Logic will show a mono-stereo version of Periscope, which doesn't really exist. To save cpu time, always choose a channel configuration of Periscope that has as many inputs as outputs.

Note that Logic also does not support effect plug-ins with more than stereo outputs.

To use Midi control and surround outputs with the Nautilus plug-ins, you must insert them in the instrument slot of an instrument object. Then select an input source from the sidechain menu to pass audio to the plug-ins inputs.



Midi Control of the Nautilus Bundle.

Version 2.0.2 of the Nautilus Bundle adds significant Midi control. Note that Logic does not allow effect plug-ins to be controlled by midi.

Deep Phase Nine:

Midi controller 1: Feedback
Midi controller 11: Low frequency limit
Midi controller 12: High frequency limit
Midi controller 13: Spread
Midi controller 14: Mix
Midi controller 15: LFO Speed
Midi controller 16: Stereo offset
Midi controller 17: LFO Shape
Midi controller 18: Number of notches

Riverrun:

Midi controller 1: Mono/Stereo width
Midi controller 11: Left needle
Midi controller 12: Right needle
Midi controller 13: Walk speed
Midi controller 14: Grain speed
Midi controller 15: Pitch
Midi controller 16: Pitch Random
Midi controller 17: Grain Speed Random
Midi controller 18: Grain Lengh

Patch change:	Envelope choice
Note on:	Positions the grain pick glass. On a 61-keys keyboard the lowest note is full left, the highest note is full right Notes outside this range wrap around.

Periscope:

Midi controller 32-63:	Fader 1-32
Midi controller 64:	Left needle
Midi controller 65:	Right needle
Note on:	Sets fader value to velocity of note. Note number 64-75 are the faders, other note values wrap around.
Note off:	Resets the fader to the position it was before note on. As the fader distribution depends on the red and yellow needles, the frequencies of the faders played by notes do not correspond to any sensible note frequencies; a note plays a corresponding fader no matter what frequency that's actually controlling.
Pitch Bend:	Shifts red and yellow needle as a pair, left for pitch down, right for pitch up.



Version History.

For clarity, the version history of the OS9, MAS-only version is omitted.

Version 2:

First OS X release. Nautilus now includes the HTDM, RTAS, VST and AU formats in addition to MAS.

Version 2.0.2:

The MAS version of Nautilus could cause Digital Performer to crash if the Motu Audio System was switched off and on.

Added Midi support to all plug-ins, with the exception of AudioUnits.

Version 2.0.3:

Added Midi control to the AudioUnit. To use a Midi controlled AudioUnit in Logic, insert it in the instrument slot of a an instrument object. Via the sidechain input you can route input from another track or bus to the plug-in. Sending Midi data to it works in the same way as you would when using an instrument.

Unfortunately, as Logic traps some controllers before passing them to the plug-in we had to renumber the controllers. The Midi Control page has the revised numbers.

Behaviour of note numbers has remained the same.

Some modifications were made to the way the plug-ins send and receive automation data. For you as user this won't make a difference, other than that the automation system works more predictable. Version 2.0.2 and lower could receive excessive parameter values from automation ramps.

Version 2.1.0:

The Pro Tools versions, HTDM and RTAS are completely rewritten, and the AudioSuite version added. They are now installed as single plug-in, and show up in your Pro Tools session where appropriate.

The AudioUnit version can now be automated in Logic.

For all plug-ins performance and automation were tweaked.